Solutions Tutorial 8

1. Evaluate the following iterated integrals

(a) $\int_1^2 \int_{1-x}^{\sqrt{x}} x^2 y \; dydx$ (b) $\int_{-1}^1 \int_{x^3}^{x+1} (3x+2y) \; dydx$ (c) $\int_0^1 \int_y^1 \frac{dxdy}{1+y^2}$

(d) $\int_0^1 \int_1^0 xy^2 dxdy$ (e) $\int_0^1 \int_0^{\pi y} \sin\left(\frac{x}{y}\right) dxdy$ (f) $\int_0^1 \int_0^x e^{xy} x^2 \; dydx$

Solution. (a) $\int_1^2 \int_{1-x}^{\sqrt{x}} x^2 y \; dydx = \frac{1}{2}\int_1^2 \Big[(xy)^2\Big]_{1-x}^{\sqrt{x}} dx = \frac{1}{2}\int_1^2 x^2(3x-1-x^2)dx = \frac{163}{120}$.

(b) $\int_{-1}^1 \int_{x^3}^{x+1} (3x+2y) \; dydx = \int_{-1}^1 \Big[3xy+y^2\Big]_{x^3}^{x+1} dx = \int_{-1}^1 \Big((3x(x+1)+(x+1)^2-3x^4-x^6\Big)dx$

$= \int_{-1}^1 \Big((3x^2+3x+(x+1)^2-3x^4-x^6\Big)dx = \Big[x^3+\frac{3}{2}x^2+\frac{1}{3}(x+1)^3-\frac{3}{5}x^5-\frac{1}{7}x^7\Big]_{-1}^1 = \frac{334}{105}$.

(c) $\int_0^1 \int_y^1 \frac{dxdy}{1+y^2} = \int_0^1 \Big[\frac{x}{1+y^2}\Big]_y^1 dy = \int_0^1 \Big[\frac{1}{1+y^2}-\frac{y}{1+y^2}\Big]_y^0 dy = \Big[\tan^{-1}y-\frac{1}{2}\ln(1+y^2)\Big]_0^1 = \frac{\pi}{4}-\frac{1}{2}\ln 2$

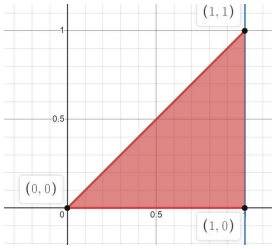
(d) $\int_0^1 \int_1^0 xy^2 dxdy = \frac{1}{2}\int_0^1 \Big[x^2y^2\Big]_1^0 dy = -\frac{1}{2}\int_0^1 y^2 dy = -\frac{1}{6}y^3\Big|_0^1 = -\frac{1}{6}$.

Alternative: $\int_0^1 \int_1^0 xy^2 dxdy = \int_1^0 \int_0^1 xy^2 dydx = \frac{1}{3}\int_1^0 \Big[xy^3\Big]_0^1 dx = \frac{1}{3}\int_1^0 x\,dx = \frac{1}{6}y^2\Big|_1^0 = -\frac{1}{6}$.

(e) $\int_0^1 \int_0^{\pi y} \sin\left(\frac{x}{y}\right) dxdy = -\int_0^1 \Big[y\cos\left(\frac{x}{y}\right)\Big]_0^{\pi y} dy = \int_0^1 2y\,dy = 1$

Note: It is not easy to integrate if we switch to $dydx$ for this question.

(f) $\int_0^1 \int_0^x e^{xy} x^2 dydx = \int_0^1 \Big[e^{xy}x\Big]_0^x dx = \int_0^1 (xe^{x^2}-x)\,dx = \frac{1}{2}(e^{x^2}-x^2)\Big|_0^1 = \frac{1}{2}(e-2)$



Remark: $\int xe^{ax}dx = \frac{e^{ax}}{a^2}(ax-1)+C$

$\int x^2e^{ax}dx = \frac{1}{a}x^2e^{ax} - \frac{2}{a}\int xe^{ax}dx$

$= \frac{1}{a}x^2e^{ax} - \frac{2}{a^2}\Big((x-1)e^{ax}\Big)+C$

$= \frac{1}{a^2}e^{ax}\Big(ax^2-2x+2\Big)+C$

2. Evaluate the following.

(a) $\iint_D x^2 y \, dxdy$, $D = \{(x,y) \in \mathbb{R}^2 : 0 \le x \le 2,\ 1 \le y \le 3\}$

(b) $\iint_D e^{2x+y} \, dxdy$, $D = \{(x,y) \in \mathbb{R}^2 : 0 \le x \le 1,\ 0 \le y \le 3\}$

(c) $\iint_D \frac{e^y}{y} \, dxdy$, $D = \{(x,y) : \sqrt{x} \le y \le 2, 0 \le x \le 4\}$

(d) $\iint_T (x-y) \, dxdy$, $T =$ triangle with vertices (0, 0), (1, 0), (2, 1)

(e) $\iint_P xy \, dxdy$, $P =$ parallelogram with vertices (0, 0), (1, 1), (1, 0), (2, 1)

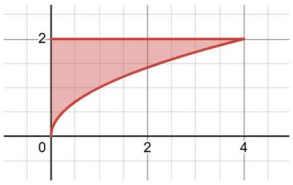
(f) $\iint_D \left(x^3 + 4y\right) dxdy$, $D =$ region bounded by curves $y = x^2$ and $y = 2x$

Solution. (a) $\int_1^3 \int_0^2 x^2 y \, dxdy = \frac{1}{3} \int_1^3 \left[x^3 y\right]_0^2 dy = \frac{1}{3} \int_1^3 8y \, dy = \frac{32}{3}$.

Alternative: $\int_1^3 \int_0^2 x^2 y \, dxdy = \int_0^2 \int_1^3 x^2 y \, dydx = \left[\frac{x^3}{3}\right]_0^2 \times \left[\frac{y^2}{2}\right]_1^3 = \frac{8}{3} \times \frac{9-1}{2} = \frac{32}{3}$.

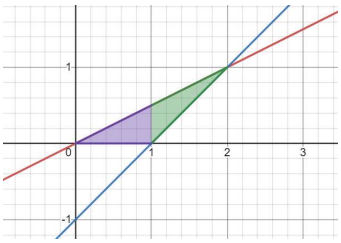
(b) $\int_0^3 \int_0^1 e^{2x} e^y dxdy = \int_0^1 e^{2x} dx \int_0^3 e^y dy = \frac{1}{2}\left[e^{2x}\right]_0^1 \times \left[e^y\right]_0^3 = \frac{1}{2}(e^2 - 1)(e^3 - 1)$.

(c)



$$\int_0^2 \int_0^{y^2} \frac{e^y}{y} \, dxdy = \int_0^2 \left[\frac{xe^y}{y}\right]_0^{y^2} dy = \int_0^2 ye^y dy = e^2 + 1$$

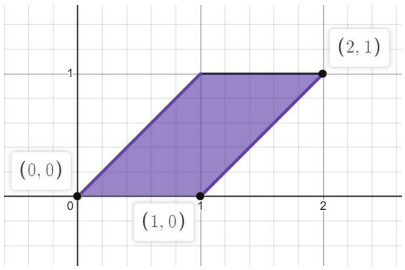
(d)



$$\int_0^1 \int_{y=0}^{y=\frac{x}{2}} (x-y) dydx + \int_1^2 \int_{y=x-1}^{y=\frac{x}{2}} (x-y) dydx$$
$$= \int_0^1 \left[xy - \frac{1}{2}y^2\right]_0^{y=\frac{x}{2}} dx + \int_1^2 \left[xy - \frac{1}{2}y^2\right]_{x-1}^{y=\frac{x}{2}} dx$$
$$= \int_0^1 \frac{3}{8}x^2 dx + \int_1^2 \frac{1}{2} - \frac{1}{8}x^2 dx = \frac{1}{8} + \frac{5}{24} = \frac{1}{3}$$

Alternative (is faster): $\int_0^1 \int_{x=2y}^{x=y+1} (x-y) \, dxdy$.

(e) We learned from (d) that now we should compute the double integral in this way:

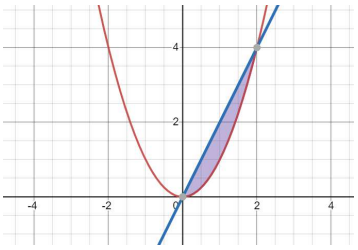


$$\int_0^1 \int_{x=y}^{x=y+1} xy\,dxdy = \int_0^1 \Big[\tfrac{1}{2}x^2 y\Big]_y^{y+1} dy = \int_0^1 \Big(y^2 + \tfrac{y}{2}\Big)dy = \tfrac{7}{12}$$

Alternative (but harder):

$$\int_0^1 \int_{y=0}^{y=x} (x-y)\,dydx + \int_1^2 \int_{y=x-1}^{y=1} (x-y)\,dydx\,.$$

(f) From the graph we see that



$$\int_0^2 \int_{y=x^2}^{y=2x} (x^3+4y)dydx = \int_0^4 \int_{x=\frac{y}{2}}^{x=\sqrt{y}} (x^3+4y)dxdy = \tfrac{32}{3}$$

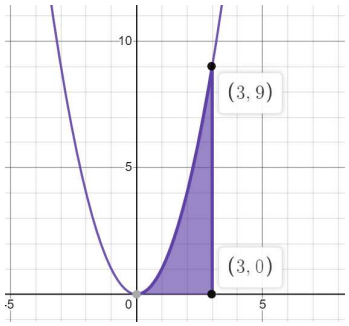
(You may choose either one to start with.)

3. Find the following integrals by first interchanging the order of integration:

(a) $\int_0^9 \int_{\sqrt{y}}^3 \sin(x^3)dxdy$ (b) $\int_0^1 \int_x^1 \frac{x}{\sqrt{x^2+y^2}}dydx$ (c) $\int_0^2 \int_{\frac{x}{2}}^1 \log(1+y^2)dydx$

(d) $\int_0^a \int_0^{a-\sqrt{a^2-y^2}} \frac{xy\ln(x+a)}{(x-a)^2}dxdy,\ a>0$ (e) $\int_0^{13} \int_{2+(1+2x)^{1/3}}^5 \frac{dydx}{(y-2)^2+y-1}$

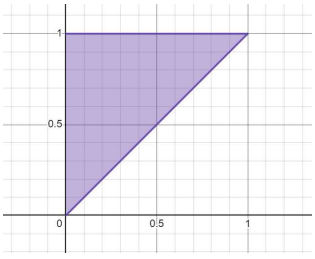
Solution. (a) $\frac{d}{dx}\cos\Big(f(x)\Big) = -f'(x)\sin\Big(f(x)\Big) \Rightarrow \int f'(x)\sin\Big(f(x)\Big)dx = -\cos\Big(f(x)\Big) + C\,.$



$$\int_0^9 \int_{\sqrt{y}}^3 \sin(x^3)\,dxdy = \int_0^3 \int_0^{x^2} \sin(x^3)\,dydx$$
$$= \int_0^3 \Big[y\sin(x^3)\Big]_0^{x^2} dx = \int_0^3 x^2\sin(x^3)\,dx$$
$$= -\tfrac{1}{3}\cos(x^3)\Big|_0^3 = \tfrac{1}{3}(1-\cos 27)$$

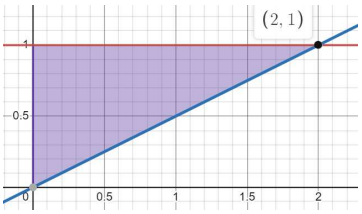
Remark: $\cos 27$ is computed in radian.

(b) Note that $\frac{d}{dx}(f(x))^n = nf'(x)(f(x))^{n-1} \Rightarrow \int f'(x)(f(x))^{n-1}dx = \frac{1}{n}(f(x))^n + C$



$$\int_0^1 \int_x^1 \frac{x}{\sqrt{x^2+y^2}}\,dydx = \int_0^1 \int_{x=0}^{x=y} \frac{x}{\sqrt{x^2+y^2}}\,dxdy$$
$$= \int_0^1 \Big[\sqrt{x^2+y^2}\Big]_0^y dy$$
$$= (\sqrt{2}-1)\int_0^1 y\,dy = \tfrac{1}{2}(\sqrt{2}-1)$$

(c) $\int \ln x\,dx = \int 1\cdot \ln x\,dx = x\ln x - \int x\cdot\left(\frac{1}{x}\right)dx = x\ln x - \int 1\,dx = x\ln x - x + C$.



$$\int_0^2\int_{x/2}^1 \log(1+y^2)\,dydx = \int_0^1\int_0^{2y} \log(1+y^2)\,dxdy$$
$$= \int_0^1 \Big[x\log(1+y^2)\Big]_0^{2y}\,dy = \int_0^1 \Big(2y\log(1+y^2)\Big)dy$$

Remark: log is ln, i.e., natural logarithm.

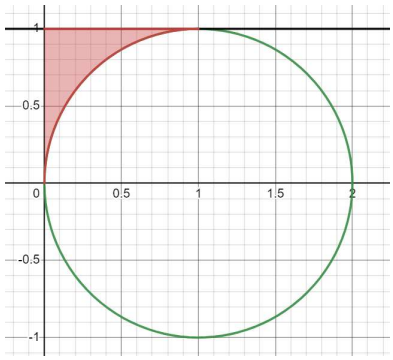
Let $u = 1+y^2 \Rightarrow du = 2ydy$. Then

$$\int_0^1 \Big(2y\log(1+y^2)\Big)dy = \int_1^2 \Big(\log(u)\Big)du = u\ln u - u\Big|_1^2 = 2\ln 2 - 1.$$

(d) $\int_0^a \int_0^{a-\sqrt{a^2-y^2}} \frac{xy\ln(x+a)}{(x-a)^2}\,dxdy,\ a>0$ Given that $a>0$, then we write $(x-a)^2 = a^2 - y^2$

or $y = \pm\sqrt{2ax-x^2}$ and see that this is a circle with centre $(a,\,0)$ and radius a.

$\sqrt{2ax-x^2} \le y \le a$ for $0 \le x \le a$, i.e.,

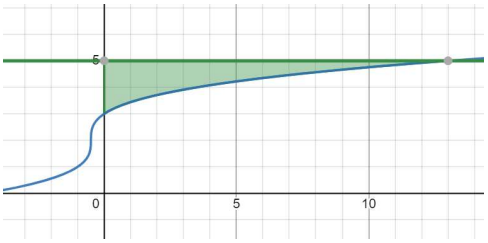


(picture drawn in the multiple of a)

$$\int_0^a \int_0^{a-\sqrt{a^2-y^2}} \frac{xy\ln(x+a)}{(x-a)^2}\,dxdy = \int_0^a \int_{\sqrt{2ax-x^2}}^a \frac{xy\ln(x+a)}{(x-a)^2}\,dydx$$
$$= \frac{1}{2}\int_0^a \left[\frac{xy\ln(x+a)}{(x-a)^2}\right]_{\sqrt{2ax-x^2}}^a dx = \frac{1}{2}\int_0^a \frac{(a^2-2ax+x^2)x\ln(x+a)}{(x-a)^2}\,dx$$
$$= \frac{1}{2}\int_0^a \frac{\cancel{(a^2-2ax+x^2)}x\ln(x+a)}{\cancel{(x-a)^2}}\,dx$$
$$= \frac{1}{2}\int_0^a x\ln(x+a)\,dx = \frac{1}{2}\left[\frac{x^2}{2}\ln(x+a)\right]_0^a - \frac{1}{2}\int_0^a \frac{x^2}{2}\cdot\frac{1}{x+a}\,dx = \frac{a^2}{4}\ln(2a) - \frac{1}{4}\int_0^a \left(x-a+\frac{a^2}{x+a}\right)dx\ *$$
$$= \frac{a^2}{4}\ln(2a) - \frac{1}{4}\left[\frac{x^2}{2} - ax + a^2\ln(x+a)\right]_0^a = \frac{a^2}{4}\ln(2a) - \frac{1}{4}\left[-\frac{a^2}{2} + a^2\ln(2a) - a^2\ln a\right] = \frac{a^2}{8}(1+2\ln a)$$

$* \ \frac{x^2}{x+a} = \frac{(x+a)^2-2ax-a^2}{x+a} = \frac{(x+a)^2-2a(x+a)+2a^2-a^2}{x+a} = x+a-2a+\frac{a^2}{x+a}$.

(e)



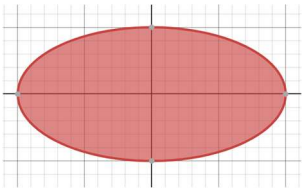
$$\int_0^{13}\int_{2+(1+2x)^{1/3}}^5 \frac{dydx}{(y-2)^2+y-1} = \int_3^5 \int_0^{\frac{1}{2}\left[(y-2)^3-1\right]} \frac{dxdy}{(y-2)^2+y-1}$$
$$= \int_3^5 \left[\frac{x}{(y-2)^2+y-1}\right]_0^{\frac{1}{2}\left[(y-2)^3-1\right]} dy = \frac{1}{2}\int_3^5 \frac{(y-2)^3-1}{(y-2)^2+y-1}\,dy$$
$$= \frac{1}{2}\int_3^5 \Big(y-3\Big)\,dy = \frac{1}{2}\left[\frac{y^2}{2} - 3y\right]_3^5 = 1$$

where $\frac{(y-2)^3-1}{(y-2)^2+y-1} = \frac{(y-2)^3-1}{(y-2)^2+(y-2)+1} = \frac{u^3-1}{u^2+u+1} = \frac{(u-1)(u^2+u+1)}{u^2+u+1} = u-1 = y-3$

4. Find the area of the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ by using double integration.

Solution. The area of the ellipse $=4\times$ area of the ellipse on the first quadrant

Let $a,b>0$.



$$A=\iint_D 1\ dydx=4\int_{x=0}^{a}\int_{y=0}^{y=b\sqrt{1-\frac{x^2}{a^2}}} 1\ dydx$$
$$=4\int_0^a y\Big|_0^{b\sqrt{1-\frac{x^2}{a^2}}}dx=4b\int_0^a\sqrt{1-\frac{x^2}{a^2}}\ dx=4b\left(\frac{a\pi}{4}\right)=ab\pi$$

Substitute $x=a\sin\theta\Rightarrow dx=a\cos\theta d\theta$:

$$\int_0^a\sqrt{1-\frac{x^2}{a^2}}\ dx=\int_0^{\frac{\pi}{2}}a\cos^2\theta\ d\theta=\frac{a}{2}\int_0^{\frac{\pi}{2}}(2\cos 2\theta+1)\ d\theta=\frac{a}{2}\Big[\sin 2\theta+\theta\Big]_0^{\frac{\pi}{2}}=\frac{a\pi}{4}.$$

5. By converting to polar coordinates, find $\iint_D\sqrt{x^2+y^2}dA$ where D is the region inside the circle $x^2+y^2=a^2$.

Solution. Let $x=r\cos\theta, y=r\sin\theta$, then $dxdy=rdrd\theta$. With the region inside the circle $x^2+y^2=a^2$, we have $0\le\theta\le 2\pi$ in $0\le r\le a$. Since $z=\sqrt{x^2+y^2}$ is symmetry, it follows that $\iint_D\sqrt{x^2+y^2}dA=\int_0^a\int_0^{2\pi}r^2d\theta dr=2\int_0^a r^2dr\int_0^{\pi}d\theta=\frac{2}{3}\pi a^3$.

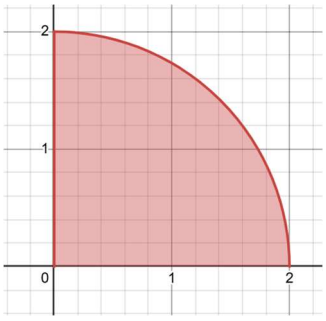
6. Sketch the region of integration for the following integrals and evaluate the integrals by using polar coordinates

(a) $\int_0^2\int_0^{\sqrt{4-y^2}}(4-x^2-y^2)^{3/2}dxdy$

(b) $\int_0^a\int_{-\sqrt{ax-x^2}}^{\sqrt{ax-x^2}}(a^2-x^2-y^2)^{1/2}dydx$, $a>0$

(c) $\int_0^1\int_x^{\sqrt{2x-x^2}}\frac{dydx}{\sqrt{x^2+y^2}}$

Solution. (a) Sketch the graph $0\le x\le\sqrt{4-y^2}$ in $0\le y\le 2$.

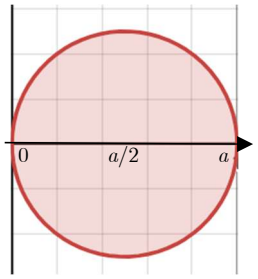


Let $x=r\cos\theta, y=r\sin\theta$.

We have $0\le\theta\le\frac{\pi}{2}$ in $0\le r\le 2$. Therefore,

$$\int_0^2\int_0^{\sqrt{4-y^2}}(4-x^2-y^2)^{3/2}dxdy=\int_0^2\int_0^{\pi/2}(4-r^2)^{3/2}rd\theta dr$$
$$=\int_0^2\frac{\pi}{2}r(4-r^2)^{3/2}dr=-\frac{\pi}{10}(4-r^2)^{5/2}\Big|_0^2=\frac{16\pi}{5}$$

(b) $\int_0^a \int_{-\sqrt{ax-x^2}}^{\sqrt{ax-x^2}} \sqrt{a^2-x^2-y^2}\,dydx = 2\int_0^a \int_0^{\sqrt{ax-x^2}} \sqrt{a^2-x^2-y^2}\,dydx$ (take the first quadrant and multiply by 2)*. Substitute $x = r\cos\theta, y = r\sin\theta$. Then $0 \le \theta \le \frac{\pi}{2}$. Note that $y \le \sqrt{ax-x^2} \Rightarrow r^2\sin^2\theta \le ar\cos\theta - r^2\cos^2\theta \Rightarrow r \le a\cos\theta$.



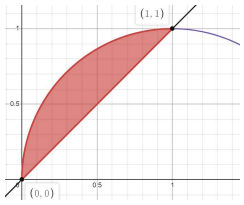
$$2\int_0^{\frac{\pi}{2}} \int_0^{a\cos\theta} r\sqrt{a^2-r^2}\,drd\theta = \tfrac{2}{3}a^3\int_0^{\frac{\pi}{2}}(1-\sin^3\theta)d\theta\,\dagger$$
$$= \tfrac{2}{3}a^3\left[\theta + \cos\theta - \tfrac{1}{3}\cos^3\theta\right]_0^{\frac{\pi}{2}}$$
$$= \tfrac{2}{3}a^3\left(\tfrac{\pi}{2} - \tfrac{2}{3}\right)$$

Remark: *We can do this because $f(x,y) = \sqrt{a^2-x^2-y^2}$ is symmetry.

† We can also obtain the answer in this way, but we need to be extra careful:

$$\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \int_0^{a\cos\theta} r\sqrt{a^2-r^2}\,drd\theta = -\tfrac{1}{3}\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\left[\sqrt{(a^2-r^2)^3}\right]_0^{a\cos\theta} d\theta = \tfrac{1}{3}\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\left(a^3 - a^3\sqrt{(1-\cos^2\theta)^3}\right)d\theta$$
$$= \tfrac{a^3}{3}\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\left(1-\left|\sin^3\theta\right|\right)d\theta = \tfrac{a^3}{3}\int_{-\frac{\pi}{2}}^{0}\left(1+\sin^3\theta\right)d\theta + \tfrac{a^3}{3}\int_0^{\frac{\pi}{2}}\left(1-\sin^3\theta\right)d\theta$$
$$= \tfrac{a^3}{3}\left[\theta - \cos\theta + \tfrac{1}{3}\cos^3\theta\right]_{-\frac{\pi}{2}}^{0} + \tfrac{a^3}{3}\left[\theta + \cos\theta - \tfrac{1}{3}\cos^3\theta\right]_0^{\frac{\pi}{2}}$$
$$= \tfrac{a^3}{3}\left(-\tfrac{2}{3} + \tfrac{\pi}{2}\right) + \tfrac{a^3}{3}\left(\tfrac{\pi}{2} - \tfrac{2}{3}\right) = \tfrac{2a^3}{3}\left(\tfrac{\pi}{2} - \tfrac{2}{3}\right)$$

(c) $\int_0^1 \int_x^{\sqrt{2x-x^2}} \frac{dydx}{\sqrt{x^2+y^2}}$. $y \le \sqrt{2x-x^2} \Rightarrow r\sin\theta \le \sqrt{2r\cos\theta - r^2\cos^2\theta} \Rightarrow r \le 2\cos\theta$

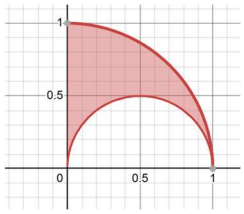


Substitute $x = r\cos\theta, y = r\sin\theta$. For $\frac{\pi}{4} \le \theta \le \frac{\pi}{2}$,

$$\int_0^1 \int_x^{\sqrt{2x-x^2}} \frac{dydx}{\sqrt{x^2+y^2}} = \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \int_0^{2\cos\theta} \frac{rdrd\theta}{r} = \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} 2\cos\theta\,d\theta = 2 - \sqrt{2}$$

7. Evaluate $\iint_D \frac{dxdy}{\sqrt{1-x^2-y^2}}$ where D is the region in the first quadrant between circles $x^2+y^2=1$ and $(x-\frac{1}{2})^2 + y^2 = \frac{1}{4}$.

Solution. Substitute $x = r\cos\theta$ and $y = r\sin\theta$, $\cos\theta \le r \le 1$ and $0 \le \theta \le \frac{\pi}{2}$.



$$\int_0^{\frac{\pi}{2}} \int_{\cos\theta}^1 \frac{rdrd\theta}{\sqrt{1-r^2}} = -\int_0^{\frac{\pi}{2}}\left[\sqrt{1-r^2}\right]_{\cos\theta}^1 d\theta = \int_0^{\frac{\pi}{2}} \sin\theta\,d\theta = -\left[\cos\theta\right]_0^{\frac{\pi}{2}} = 1$$

Alternative: Let $D_1 = \{(x,y) : 0 \le x^2 + y^2 \le 1, x, y \ge 0\}$, and $D_2 = \{(x,y) : 0 \le (x - \frac{1}{2})^2 + y^2 \le \frac{1}{4}, x, y \ge 0\}$. Then

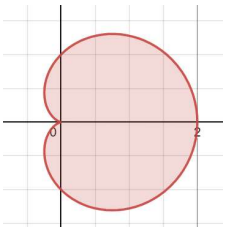
$$\iint_{D_1} \frac{dxdy}{\sqrt{1-x^2-y^2}} = \int_0^{\frac{\pi}{2}} \int_0^1 \frac{rdrd\theta}{\sqrt{1-r^2}} = -\frac{\pi}{2}\left[\sqrt{1-r^2}\right]_0^1 = \frac{\pi}{2}$$

$$\iint_{D_2} \frac{dxdy}{\sqrt{1-x^2-y^2}} = \int_0^{\frac{\pi}{2}} \int_0^{\cos\theta} \frac{rdrd\theta}{\sqrt{1-r^2}} = -\int_0^{\frac{\pi}{2}} \left[\sqrt{1-r^2}\right]_0^{\cos\theta} d\theta = \left[\cos\theta + \theta\right]_0^{\frac{\pi}{2}} = \frac{\pi}{2} - 1$$

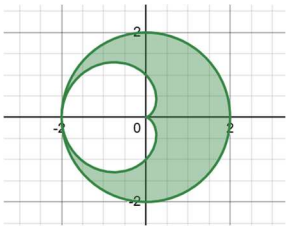
Hence, $\iint_D \frac{dxdy}{\sqrt{1-x^2-y^2}} = \iint_{D_1} \frac{dxdy}{\sqrt{1-x^2-y^2}} - \iint_{D_2} \frac{dxdy}{\sqrt{1-x^2-y^2}} = \frac{\pi}{2} - \left(\frac{\pi}{2} - 1\right) = 1$.

8. Sketch the curve with polar equation $r = 1 + \cos\theta$ for $-\pi \le \theta \le \pi$. Find the area enclosed within the curve.

Solution. Since, $r = 1 + \cos\theta$ for $-\pi \le \theta \le \pi$, it follows that the area of the cardioid is equal to $\iint_D 1\, dxdy = 2\int_0^{\pi} \int_0^{1+\cos\theta} rdrd\theta = \int_0^{\pi} (1+\cos\theta)^2\, d\theta = \frac{3}{2}\pi$.



Q8



Q9

9. Find the area inside the circle $r = 2$ but outside of the cardioid $r = 1 - \cos\theta$.

Solution.

$$A = 2\int_0^{\pi} \int_{1-\cos\theta}^{2} rdrd\theta$$
$$= \int_0^{\pi} 4 - (1-\cos\theta)^2 d\theta$$
$$= \int_0^{\pi} (3 + 2\cos\theta - \cos^2\theta) d\theta$$
$$= \left[3\theta + 2\sin\theta - \tfrac{1}{2}(\theta + \tfrac{1}{2}\sin(2\theta))\right]_0^{\pi}$$
$$= \frac{5}{2}\pi$$

or

$$A = \int_0^{2\pi} \int_0^{2} rdrd\theta - 2\int_0^{\pi} \int_0^{1-\cos\theta} rdrd\theta$$
$$= 4\pi - \int_0^{\pi} (1-\cos\theta)^2\, d\theta$$
$$= 4\pi - \int_0^{\pi} \left(\tfrac{3}{2} - 2\cos\theta + \tfrac{1}{2}\cos 2\theta\right) d\theta$$
$$= 4\pi - \left[\tfrac{3}{2}\theta - 2\sin\theta + \tfrac{1}{4}\sin 2\theta\right]_0^{\pi}$$
$$= 4\pi - \tfrac{3}{2}\pi = \tfrac{5}{2}\pi$$

Extra Exercises

For Q10 to 17: Evaluate the integrals.

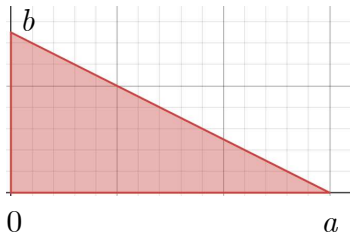
10. $$\int_0^2 \int_0^{\sqrt{4-y^2}} dxdy = \int_0^2 \Big[x\Big]_0^{\sqrt{4-y^2}} dy = \int_0^2 \sqrt{4-y^2}\,dy = \int_0^{\pi/2} \sqrt{4-4\sin^2\theta}\cdot 2\cos\theta d\theta$$
$$= 4\int_0^{\pi/2} \cos^2\theta d\theta = 2\int_0^{\pi/2} (1+\cos 2\theta)d\theta = 2\Big[\theta + \tfrac{1}{2}\sin 2\theta\Big]_0^{\pi/2} = \pi$$

Note that the domain of the integral is the first quadrant part of the circular disk of radius 2 centred at (0, 0). So the answer is π.

11. $$\int_0^2 \int_0^3 12x^2y^3 dxdy = \int_0^2 \Big[4x^3y^3\Big]_0^3 dy = \int_0^2 4\left(3^3\right)y^3 dy = \Big[27y^4\Big]_0^2 = 27\left(2^4\right) = 432.$$

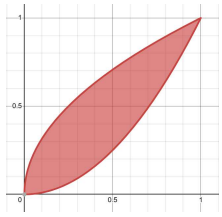
12. $$\int_0^3 \int_0^2 12x^2y^3 dydx = 432$$ (same answer as Q11).

13. $\iint_T (x-3y)dxdy$, where T is the triangle with vertices (0,0), (a,0), (0,b).



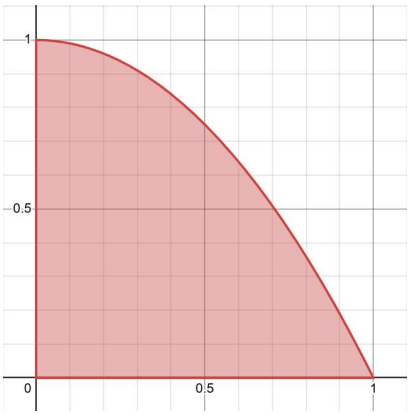
$$\iint_T (x-3y)dxdy = \int_0^a \int_{y=0}^{y=b-\frac{b}{a}x} (x-3y)\,dydx$$
$$= \int_0^a \Big[xy - \tfrac{3}{2}y^2\Big]_{y=0}^{y=b-\frac{b}{a}x} dx$$
$$= \int_0^a x\left(b-\tfrac{b}{a}x\right) - \tfrac{3}{2}\left(b-\tfrac{b}{a}x\right)^2 dx$$
$$= \tfrac{b}{2a^2}\int_0^a 2a^2x - 2ax^2 - 3b\left(a-x\right)^2 dx$$
$$= \tfrac{b}{2a^2}\Big[a^2x^2 - \tfrac{2}{3}ax^3 + b\left(a-x\right)^3\Big]_0^a$$
$$= \tfrac{b}{2a^2}\left(a^4 - \tfrac{2}{3}a^4 - a^3b\right) = \tfrac{b}{2}\left(\tfrac{1}{3}a^2 - ab\right)$$
$$= \tfrac{ab(a-3b)}{6}$$

14. $\iint_R xy^2 dxdy$, where R is the first quadrant bounded by $y = x^2$ and $x = y^2$.



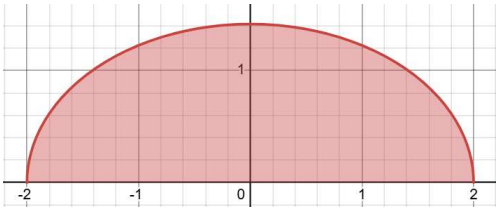
$$\iint_R xy^2 dxdy = \int_{y=0}^{y=1} \int_{x=y^2}^{x=\sqrt{y}} xy^2\,dxdy = \tfrac{1}{2}\int_{y=0}^{y=1} \Big[x^2y^2\Big]_{x=y^2}^{x=\sqrt{y}} dy = \tfrac{1}{2}\int_{y=0}^{y=1} \left(y^3 - y^6\right) dy$$
$$= \tfrac{1}{2}\Big[\tfrac{1}{4}y^4 - \tfrac{1}{7}y^7\Big]_0^1 = \tfrac{1}{2}\left(\tfrac{1}{4} - \tfrac{1}{7}\right) = \tfrac{3}{56}$$

15. $\iint_D x\cos y dxdy$, D is the first quadrant bounded by the axes and $y = 1 - x^2$.



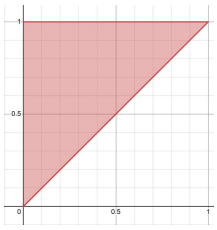
$$\begin{aligned}\iint_D x\cos y dxdy &= \int_{y=0}^{y=1}\int_{x=0}^{x=\sqrt{1-y}} x\cos y\ dxdy \\ &= \int_{y=0}^{y=1}\left[\tfrac{1}{2}x^2\cos y\right]_0^{\sqrt{1-y}} dy \\ &= \tfrac{1}{2}\int_{y=0}^{y=1}(1-y)\cos y\ dy \\ &= \tfrac{1}{2}\Big[(1-y)\sin y\Big]_0^1 - \tfrac{1}{2}\int_{y=0}^{y=1} -\sin y\ dy \\ &= \tfrac{1}{2}\Big[-\cos y\Big]_0^1 = \tfrac{1}{2}\Big(1-\cos 1\Big)\end{aligned}$$

16. $\int_0^{\sqrt{2}} dy\int_{-\sqrt{4-2y^2}}^{\sqrt{4-2y^2}} dx\ y$ (this is equivalent to $\int_0^{\sqrt{2}}\int_{-\sqrt{4-2y^2}}^{\sqrt{4-2y^2}} y\ dx\ dy$)



$$\begin{aligned}\int_0^{\sqrt{2}}\int_{-\sqrt{4-2y^2}}^{\sqrt{4-2y^2}} y\ dxdy &= 2\int_0^{\sqrt{2}}\int_0^{\sqrt{4-2y^2}} y\ dxdy \\ &= 2\int_0^{\sqrt{2}} y\sqrt{4-2y^2}\,dy \\ &= \left[-\tfrac{1}{3}\Big(4-2y^2\Big)^{\frac{3}{2}}\right]_0^{\sqrt{2}} \\ &= -\tfrac{1}{3}\Big(4-4\Big)^{\frac{3}{2}} + \tfrac{1}{3}\Big(4\Big)^{\frac{3}{2}} = \tfrac{8}{3}\end{aligned}$$

17. Sketch the domain of $\int_0^1\int_x^1 e^{\frac{x}{y}} dydx$ and evaluate the integral.



$$\begin{aligned}\int_0^1\int_x^1 e^{\frac{x}{y}} dydx &= \int_0^1\int_0^y e^{\frac{x}{y}} dxdy = \int_0^1\left[ye^{\frac{x}{y}}\right]_0^y dy \\ &= \Big(e-1\Big)\int_0^1 ydy = \tfrac{e-1}{2}\end{aligned}$$

18. Let $I = \iint_R f(x,y)\ dA = \int_1^{\sqrt{2}}\int_{\frac{1}{y}}^{\sqrt{y}} f(x,y)\ dxdy + \int_{\sqrt{2}}^4\int_{\frac{y}{2}}^{\sqrt{y}} f(x,y)\ dxdy$.

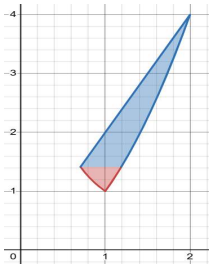
(a) Sketch the region R.

(b) Rewrite I by reversing the order of integration.

(c) Compute I when $f(x,y) = \frac{x}{y}$.

Solution.

(a)



(b) $I = \int_{\frac{\sqrt{2}}{2}}^{1}\int_{1/x}^{2x} f(x,y)\,dydx + \int_{1}^{2}\int_{x^2}^{2x} f(x,y)\,dydx$

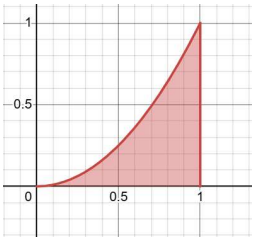
(c) $I = \int_{1}^{\sqrt{2}}\int_{\frac{1}{y}}^{\sqrt{y}} \frac{x}{y}\,dxdy + \int_{\sqrt{2}}^{4}\int_{\frac{y}{2}}^{\sqrt{y}} \frac{x}{y}\,dxdy = \int_{1}^{\sqrt{2}}\left[\frac{x^2}{2y}\right]_{\frac{1}{y}}^{\sqrt{y}}dy + \int_{\sqrt{2}}^{4}\left[\frac{x^2}{2y}\right]_{\frac{y}{2}}^{\sqrt{y}}dy$

$= \int_{1}^{\sqrt{2}}\left(\frac{1}{2} - \frac{1}{2y^3}\right)dy + \int_{\sqrt{2}}^{4}\left(\frac{1}{2} - \frac{y}{8}\right)dy = \left[\frac{1}{2}y + \frac{1}{4y^2}\right]_{1}^{\sqrt{2}} + \left[\frac{1}{2}y - \frac{y^2}{16}\right]_{\sqrt{2}}^{4}$

$= \frac{1}{2}\sqrt{2} + \frac{1}{8} - \frac{1}{2} - \frac{1}{4} + \frac{1}{2}(4) - \frac{4^2}{16} - \frac{1}{2}\sqrt{2} + \frac{2}{16} = \frac{1}{2}$

19. Sketch the domain of $\int_{0}^{1}\int_{\sqrt{y}}^{1} \frac{\sin(\pi x^2)}{x}\,dxdy$ and evaluate the integral.

Solution.



$\int_{0}^{1}\int_{\sqrt{y}}^{1} \frac{\sin(\pi x^2)}{x}\,dxdy = \int_{x=0}^{x=1}\int_{y=0}^{y=x^2} \frac{\sin(\pi x^2)}{x}\,dydx$

$= \int_{x=0}^{x=1} x\sin(\pi x^2)\,dx = -\frac{1}{2\pi}\cos(\pi x^2)\Big|_{0}^{1} = \frac{1}{\pi}$

20. Find the volume of the solid bounded above by the surface $z = f(x,y) = e^{-x^2}$, below by the plane $z = 0$ and over the triangle in the xy-plane formed by lines $x = 1$, $y = 0$ and $y = x$.

Solution. $V = \int_{0}^{1}\int_{0}^{x} e^{-x^2}\,dydx = \int_{0}^{1} xe^{-x^2}\,dx = \frac{1}{2}\left(1 - e^{-1}\right)$.

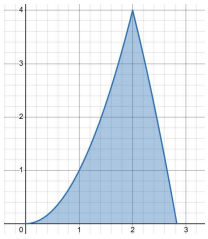
21. Let $I = \int_{0}^{4}\int_{\sqrt{y}}^{\sqrt{8-y}} f(x,y)\,dxdy$.

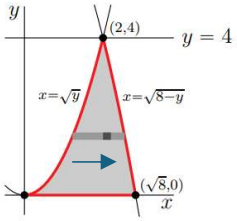
(a) Sketch the domain of I.

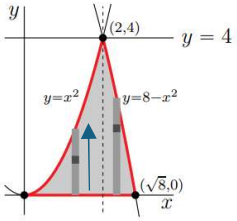
(b) Rewrite I by reversing the order of integration.

(c) Compute I when $f(x,y) = \frac{1}{(1+y)^2}$.

Solution. (a)







(b) $I = \int_0^4 \int_{\sqrt{y}}^{\sqrt{8-y}} f(x,y)\,dxdy = \int_0^2 \int_{y=0}^{y=x^2} f(x,y)\,dydx + \int_2^{2\sqrt{2}} \int_{y=0}^{y=8-x^2} f(x,y)\,dydx$

(c) $I = \int_0^2 \int_{y=0}^{y=x^2} \frac{1}{(1+y)^2}\,dydx + \int_2^{2\sqrt{2}} \int_{y=0}^{y=8-x^2} \frac{1}{(1+y)^2}\,dydx$

$= -\int_0^2 \left[\frac{1}{1+y}\right]_0^{x^2} dx - \int_2^{2\sqrt{2}} \left[\frac{1}{1+y}\right]_0^{8-x^2} dx$

$= \int_0^2 \left(1 - \frac{1}{1+x^2}\right) dx + \int_2^{2\sqrt{2}} \left(1 - \frac{1}{9-x^2}\right) dx \qquad \because \frac{1}{1+x^2} \equiv \tan^{-1} x + C$

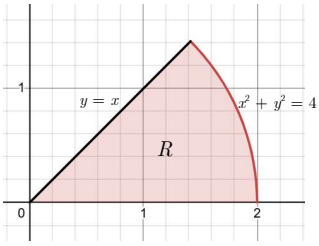
$= 2\sqrt{2} - \tan^{-1} x\Big|_0^2 - \frac{1}{6}\int_2^{2\sqrt{2}} \left(\frac{1}{3+x} + \frac{1}{3-x}\right) dx \qquad \because \frac{1}{9-x^2} \equiv \frac{1}{6}\left(\frac{1}{3+x} + \frac{1}{3-x}\right)$

$= 2\sqrt{2} - \tan^{-1} 2 - \frac{1}{6}\ln\frac{3+x}{3-x}\Big|_2^{2\sqrt{2}}$

$= 2\sqrt{2} - \tan^{-1} 2 - \frac{1}{6}\left(\ln\frac{3+2\sqrt{2}}{3-2\sqrt{2}} - \ln 5\right)$

22. For each of the region R sketched below, express $\iint_R f(x,y)\,dxdy$ as an iterated integral in polar coordinates in two different ways.

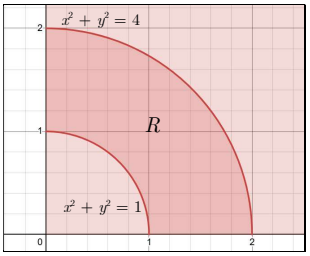
(a)



$R = \left\{(x,y) : 0 \le x^2 + y^2 \le 4, 0 \le y \le x\right\}$

$= \left\{(r\cos\theta, r\sin\theta) : 0 \le r \le 2, 0 \le \theta \le \frac{\pi}{4}\right\}$

(b)



$R = \left\{(x,y) : 1 \le x^2 + y^2 \le 4, x, y \ge 0\right\}$

$= \left\{(r\cos\theta, r\sin\theta) : 1 \le r \le 2, 0 \le \theta \le \frac{\pi}{2}\right\}$

23. Use polar coordinates to evaluate each of the following integrals.

(a) $\iint_S (x+y)\,dxdy$, where S is the region in the first quadrant lying inside the disc $x^2+y^2 \le a^2$ and under the line $y=\sqrt{3}x$.

(b) $\iint_S x\,dxdy$, where S is the disc segment $x^2+y^2 \le 2,\ x \ge 1$.

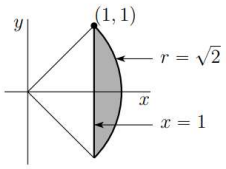
(c) $\iint_T (x^2+y^2)\,dxdy$, where T is the triangle with vertices (0, 0), (1, 0) and (1, 1).

(d) $\iint_{x^2+y^2\le 1} \ln(x^2+y^2)\,dxdy$.

Solution. (a) Since $\tan\theta = \frac{y}{x} = \sqrt{3}$, it follows that $\theta = \frac{\pi}{3}$.

$$\begin{aligned}\iint_S (x+y)\,dxdy &= \int_0^{\frac{\pi}{3}}\int_0^a r(\cos\theta+\sin\theta)\,rdrd\theta \\ &= \int_0^a r^2\,dr \int_0^{\frac{\pi}{3}} (\cos\theta+\sin\theta)\,d\theta \\ &= \tfrac{a^3}{3}\Big[\sin\theta-\cos\theta\Big]_0^{\frac{\pi}{3}} = \tfrac{a^3}{6}\Big(\sqrt{3}+1\Big)\end{aligned}$$

(b)



$x^2+y^2 \le 2 \Rightarrow r^2 \le 2$

$x \ge 1 \Rightarrow r\cos\theta \ge 1$

$\Rightarrow \frac{1}{\cos\theta} \le r \le \sqrt{2}$

$$\begin{aligned}\iint_S x\,dxdy &= 2\int_0^{\frac{\pi}{4}}\int_{\frac{1}{\cos\theta}}^{\sqrt{2}} r\cos\theta\,rdrd\theta \\ &= \tfrac{2}{3}\int_0^{\frac{\pi}{4}}\Big(2\sqrt{2}\cos\theta-\sec^2\theta\Big)d\theta \\ &= \tfrac{2}{3}\Big[2\sqrt{2}\sin\theta-\tan\theta\Big]_0^{\frac{\pi}{4}} = \tfrac{2}{3}\end{aligned}$$

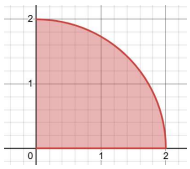
(c) Similarly to (b), $0 \le \theta \le \frac{\pi}{4}$, $0 \le r \le \frac{1}{\cos\theta}$:

$$\iint_T (x^2+y^2)\,dxdy = \int_0^{\frac{\pi}{4}}\int_0^{\frac{1}{\cos\theta}} r^3 drd\theta = \int_0^{\frac{\pi}{4}}\Big[\tfrac{r^4}{4}\Big]_0^{\frac{1}{\cos\theta}}d\theta$$

$$= \tfrac{1}{4}\int_0^{\frac{\pi}{4}} \sec^2\theta(1+\tan^2\theta)d\theta = \tfrac{1}{4}\Big[\tan\theta+\tfrac{1}{3}\tan^3\theta\Big]_0^{\frac{\pi}{4}} = \tfrac{1}{3}$$

(d) $$\begin{aligned}\iint_{x^2+y^2\le 1} \ln(x^2+y^2)\,dxdy &= \int_{r=0}^{r=1}\int_{\theta=0}^{\theta=2\pi} \ln(r^2)rd\theta dr \\ &= 2(2\pi)\int_{r=0}^{r=1} r\ln r\,dr = 4\pi\left(\tfrac{r^2}{2}\ln r\Big|_0^1 - \int_0^1 \tfrac{r^2}{2}\tfrac{1}{r}\,dr\right) \\ &= 2\pi\left(-\lim_{r\to 0^+} r^2\ln r - \tfrac{1}{2}\right) \\ &= -\pi \qquad \because \lim_{r\to 0^+}\tfrac{\ln r}{1/r^2} = \lim_{r\to 0^+}\tfrac{1/r}{-2/r^3} = \lim_{r\to 0^+}\tfrac{r^2}{-2} = 0\end{aligned}$$

24. Evaluate the iterated double integral



$$\int_{\theta=0}^{\theta=\frac{\pi}{2}}\int_{r=0}^{r=2} r^3 r\,dr\,d\theta = \frac{\pi}{2}\int_0^2 r^4\,dr = \frac{\pi}{2}\left(\frac{2^5}{5}\right) = \frac{16}{5}\pi$$

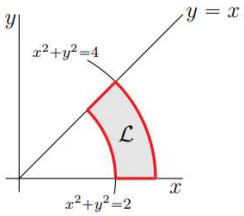
25. (a) Sketch the region L (in the first quadrant of the xy-plane) with boundary curves $x^2+y^2=2$, $x^2+y^2=4$, $y=x$, $y=0$. The mass of a thin lamina with a density function $\rho(x,y)$ over the region L is given by $M=\iint_L \rho(x,y)\,dA$.

(b) Find an expression for M as an integral in polar coordinates.

(c) Find M when $\rho(x,y)=\frac{2xy}{x^2+y^2}$.

Solution.

(a)



(b) $$M=\iint_L \rho(x,y)\,dA$$
$$=\int_0^{\frac{\pi}{4}}\int_{\sqrt{2}}^{2} r\rho(r\cos\theta, r\sin\theta)\,dr\,d\theta$$

(c) $$M=\int_0^{\frac{\pi}{4}}\int_{\sqrt{2}}^{2} r\frac{2r^2\cos\theta\sin\theta}{r^2}\,dr\,d\theta = \int_0^{\frac{\pi}{4}}\cos\theta\sin\theta\,d\theta\int_{\sqrt{2}}^{2}2r\,dr$$
$$=\left[-\frac{1}{4}\cos 2\theta\right]_0^{\frac{\pi}{4}}\left[r^2\right]_{\sqrt{2}}^{2} = \frac{1}{4}\left(4-2\right)=\frac{1}{2}$$

Summary

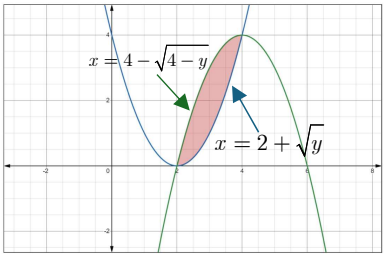
Type of integral \ Finding	Length	Area*	Volume*
Single integral $f(x):\mathbb{R}\to\mathbb{R}$	$\int_a^b 1\,dx$	$\int_a^b f(x)\,dx$	$\pi\int_a^b \big(f(x)\big)^2\,dx$
Double integral $f(x,y):\mathbb{R}^2\to\mathbb{R}$		$\iint_D 1\,dxdy$	$\iint_D f(x,y)\,dxdy$
Triple integral $f(x,y,z):\mathbb{R}^3\to\mathbb{R}$			$\iiint_D 1\,dxdydz$

*Each area and volume formula are based on different generated model/shape.

Past test/final exam questions

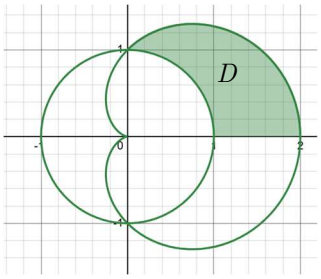
26. Let $R = \left\{(x,y) : (x-2)^2 \le y \le -(x-4)^2 + 4,\ 2 \le x \le 4\right\}$. Sketch the region R. Express the integral $\iint_R f(x,y)\, dydx$ as an integral with the order of integration interchanged. *Note: You are not required to solve the integral.*

Solution. $\int_2^4 \int_{(x-2)^2}^{-(x-4)^2+4} f(x,y)dydx = \int_0^4 \int_{4-\sqrt{4-y}}^{2+\sqrt{y}} f(x,y)dxdy$



27. Let D be the region outside of the circle $x^2 + y^2 = 1$, $x, y \in \mathbb{R}$ and inside the cardioid $r = 1 + \cos\theta$, $\theta \in [0, 2\pi]$, and D is in the first quadrant. Sketch the region D. By using a suitable substitution, evaluate $\iint_D \frac{y}{\sqrt{x^2+y^2}}\, dxdy$.

Solution.



Let $I = \iint_D \frac{y}{\sqrt{x^2+y^2}}\, dxdy$. Let $x = r\cos\theta$ and $y = r\sin\theta$. Then

$$J = \begin{vmatrix} x_r & x_\theta \\ y_r & y_\theta \end{vmatrix} = \begin{vmatrix} \cos\theta & -r\sin\theta \\ \sin\theta & r\cos\theta \end{vmatrix} = r\cos^2\theta + r\sin^2\theta = r.$$

$$\begin{aligned} I &= \int_0^{\frac{\pi}{2}} \int_1^{1+\cos\theta} r\sin\theta\, dr d\theta \\ &= \frac{1}{2}\int_0^{\frac{\pi}{2}} \left[r^2\sin\theta\right]_1^{1+\cos\theta} d\theta \\ &= \frac{1}{2}\int_0^{\frac{\pi}{2}} \left[(1+\cos\theta)^2\sin\theta - \sin\theta\right] d\theta \\ &= \frac{1}{2}\left[-\frac{(1+\cos\theta)^3}{3} + \cos\theta\right]_0^{\frac{\pi}{2}} = \frac{2}{3} \end{aligned}$$

28. Prove that $\int_0^\infty \int_0^\infty e^{-x^2-y^2}\,dxdy = \frac{\pi}{4}$. Hence, show that $\int_0^\infty e^{-x^2}\,dx = \frac{\sqrt{\pi}}{2}$.

Solution. Let $x = r\cos\theta, y = r\sin\theta$. $0 \le x, y < \infty \Rightarrow 0 \le \theta \le \frac{\pi}{2}, r \ge 0$. Note that

$$\left(\int_0^\infty e^{-x^2}dx\right)\left(\int_0^\infty e^{-y^2}dy\right) = \int_0^\infty\int_0^\infty e^{-x^2-y^2}\,dxdy = \int_0^\infty\int_0^{\frac{\pi}{2}} e^{-r^2}\,rd\theta dr\,,$$

where the righthand side can be computed as

$$\int_0^\infty \left[\theta r e^{-r^2}\right]_0^{\frac{\pi}{2}} dr = \frac{\pi}{2}\int_0^\infty re^{-r^2}dr = -\frac{\pi}{4}\lim_{b\to\infty}\left[e^{-r^2}\right]_0^b = -\frac{\pi}{4}\lim_{b\to\infty}\left(e^{-b^2}-1\right) = \frac{\pi}{4}.$$

Thus, $\int_0^\infty e^{-x^2}dx = \frac{\sqrt{\pi}}{2} \because e^{-x^2} > 0 \forall x \in \mathbb{R}$.

29. Determine the volume of the 3-dimensional region D inside the cylinder $x^2 + y^2 = 4$, bounded above by the surface $z = 2x^2 + 2y^2$ and below by the coordinate plane $z = 0$.

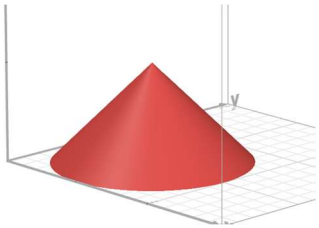
Solution. $V = \iint_D (2x^2 + 2y^2)dxdy$. Take polar coordinates, we have

$$V = \int_0^{2\pi}\int_0^2 2r^3 drd\theta = 4\pi\left[\frac{r^4}{4}\right]_0^2 = 16\pi\,.$$

30. Use a suitable substitution to evaluate the volume of a solid that is bounded below by the xy-plane and bounded above by $z = 1 - \sqrt{(x-1)^2 + (y-1)^2}$. Show all your working in details including the Jacobian.

Solution. Let $x - 1 = r\cos\theta$, $y - 1 = r\sin\theta$ $\quad J = \begin{vmatrix} x_r & x_\theta \\ y_r & y_\theta \end{vmatrix} = \begin{vmatrix} \cos\theta & \sin\theta \\ -r\sin\theta & r\cos\theta \end{vmatrix} = r$. So,

$$V = \int_0^{2\pi}\int_0^1 (1-r)rdrd\theta = 2\pi\left[\frac{r^2}{2} - \frac{r^3}{3}\right]_0^1 = \frac{\pi}{3}.$$

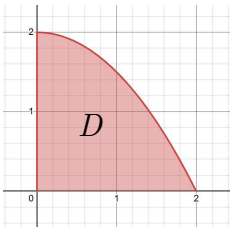


Alternative: $V = \int_0^{2\pi}\int_0^1\int_0^{1-r} rdzdrd\theta$

Check: Volume of a cone with radius 1 and height 1 is $\frac{1}{3}\pi r^2 h = \frac{\pi}{3}$.

31. Let $I = \iint_D \frac{1}{\sqrt{2y-y^2}}\,dA$, where D is the region in the first quadrant bounded by $x^2 = 4-2y$. Sketch the region D and evaluate I.

Solution.



$$\iint_D \frac{1}{\sqrt{2y-y^2}}\,dA = \int_0^2 \int_0^{\sqrt{4-2y}} \frac{1}{\sqrt{2y-y^2}}\,dxdy = \int_0^2 \left[\frac{x}{\sqrt{2y-y^2}}\right]_0^{\sqrt{4-2y}} dy = \int_0^2 \frac{\sqrt{4-2y}}{\sqrt{2y-y^2}}\,dy$$
$$= \sqrt{2}\int_0^2 \frac{\sqrt{2-y}}{\sqrt{y}\sqrt{2-y}}\,dy = \sqrt{2}\int_0^2 \frac{1}{\sqrt{y}}\,dy = 2\sqrt{2}\left[\sqrt{y}\right]_0^2 = 4$$

32. Calculate $\iint_R \frac{\sin x}{x}\,dA$, where $R \subset \mathbb{R}^2$ is the triangle bounded by the x-axis, the line $y = x$ and the line $x = \pi$.

Solution. $\iint_R \frac{\sin x}{x}\,dA = \int_{x=0}^{x=\pi} \int_{y=0}^{y=x} \frac{\sin x}{x}\,dydx = \int_{x=0}^{x=\pi} \sin x\,dx = -\cos x\Big|_0^{\pi} = 1-(-1) = 2$.

33. Let $D = \left\{(x,y,z) : 1 \le x^2+y^2 \le 4, 0 \le z \le \sqrt{x^2+y^2}\right\} \subset \mathbb{R}^3$. Find the volume of D.

Solution. $V = \int_0^{2\pi} \int_1^2 r^2 dr d\theta = 2\pi\left[\frac{1}{3}r^3\right]_1^2 = \frac{14}{3}\pi$.